

Box Transpose: In-Place N-Dimensional Permutation on GPUs

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Abstract

[PLACEHOLDER] Abstract. One paragraph: the problem, the idea, the headline result.

1 Introduction

[PLACEHOLDER] Introduction.

2 Background

[PLACEHOLDER] Permutation as a memory-movement problem; why out-of-place copies are the thing to avoid.

3 Design

3.1 Adjacent box swaps

[PLACEHOLDER] A box is a run of adjacent axes treated as one unit. A swap exchanges two adjacent boxes, and is carried out as an in-place transpose of a logical $rows \times cols$ matrix, batched over D_{pre} and vectorised over D_{post} .

3.2 Planning by shortest path

[PLACEHOLDER] The search over axis orderings, the edge cost model, and what τ and D_{post}^* prune.

3.3 Cost model

[PLACEHOLDER] Derivation of the per-step cost and how μ is calibrated.

4 Implementation

[PLACEHOLDER] The cycle-following kernel: scalar and float4 paths, the grid mapping, and the auxiliary cycle arrays.

5 Evaluation

[PLACEHOLDER] Setup, baselines, and results.

6 Related Work

[PLACEHOLDER] Related work.

7 Conclusion

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References